EVALUATION REPORT: ROUTER AUTO-SELECTION VS HEAVY-ONLY LLM

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1. EXECUTIVE SUMMARY & EFFICIENCY VERDICT

Verdict: Outstanding Efficiency: Router saved 71.5% inference latency and offloaded 100% of queries away from heavy VRAM.

Efficiency Score: 82.9 / 100

Heavy Baseline Model: llama3.1:8b

2. LATENCY & THROUGHPUT COMPARISON

â¢ Router Mean Latency: 2.56 s (p50: 2.46 s)

â¢ Heavy-Only Mean Latency: 8.97 s (p50: 6.40 s)

â¢ Latency Reduction: 71.5% efficiency gain

â¢ Average Tokens/Sec: Router 138.3 tok/s vs Heavy 18.1 tok/s

3. TOKEN USAGE & DIFFERENCE

â¢ Router Tokens: 715 total (356 prompt + 359 completion)

â¢ Heavy Tokens: 912 total (166 prompt + 746 completion)

â¢ Token Usage Difference: Router used 197 fewer tokens (21.6% difference)

4. ROUTER TIER DISTRIBUTION

â¢ Light Tier: 10 (100%)

â¢ Code Tier: 0 (0%)

â¢ Heavy Tier: 0 (0%)

5. DETAILED TEST CASE BREAKDOWN (Top Cases)

Test ID	Router Tier	R-Lat(ms)	H-Lat(ms)	Saved(ms)	Eff %	Tok Diff
eval-001	light	3211	7149	3939	55%	7
eval-002	light	3598	31846	28248	89%	202
eval-003	light	2226	6692	4466	67%	5
eval-004	light	2406	4758	2352	49%	-19
eval-005	light	2204	5609	3405	61%	-9
eval-006	light	2490	4640	2150	46%	-30
eval-007	light	2492	12368	9876	80%	68
eval-008	light	2385	5239	2854	54%	-19
eval-009	light	2465	4994	2529	51%	-19
eval-010	light	2130	6404	4274	67%	11